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Adjuvant Tislelizumab, Lenvatinib, and Capecitabine for Resected Biliary Tract Cancer: A Prospective Phase II Trial

Short title: Adjuvant immuno-target-chemotherapy for BTC

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Abstract

Background: This study aims to assess the efficacy and safety of a triplet adjuvant regimen consisting of tislelizumab, lenvatinib, and capecitabine following radical resection in patients with biliary tract cancer (BTC).

Methods: In this open-label, single-arm trial, patients with histologically confirmed BTC who had undergone curative resection were enrolled to receive adjuvant therapy within 4–16 weeks postoperatively. The treatment regimen consisted of tislelizumab (200 mg), lenvatinib (8 mg), and capecitabine (1250 mg/m²). The primary endpoint was the 1-year disease-free survival (DFS) rate. Secondary endpoints included median DFS, 2-year DFS rate, and 1- and 3-year overall survival (OS) rates, as well as safety profiles. Exploratory analyses were conducted to evaluate genomic alterations and prognostic biomarkers associated with clinical outcomes.

Results: Between February 24 and December 31, 2022, a total of 50 patients underwent treatment. As of the data cutoff (April 30, 2025), the median follow-up duration was 29.3 months (95% CI: 27.2–30.4). Intrahepatic cholangiocarcinoma constituted 78% of cases, with TNM stage III or IV observed in 40% of patients and poorly differentiated tumors identified in another 40%. The median DFS was calculated at 12.7 months (95% CI: 6.4–19.0), with DFS rates being reported as 55.5% (95% CI: 51.4%–57.6%) at one year and 30.8% (95% CI: 28.6%–32.9%) at two years, respectively. Median OS was not reached; however, the one-year OS rate stood at an impressive figure of 93.8% (95% CI: 91.7%-95.9%) while the three-year OS rate declined to 54.4% (95% CI: 50.0%–58.4%). By the exploratory analyses, *FSIP2* mutation was associated shorter DFS (P<0.001). Treatment-related grade 3 adverse events occurred in 20% of patients, with no treatment-related deaths or grade ≥ 4 toxicities reported.

Conclusions: Adjuvant tislelizumab, lenvatinib, and capecitabine demonstrated clinical activity and tolerable safety in patients with resected BTC. Despite not meeting the primary endpoint, continued

follow-up and further evaluation are warranted to better define the potential role of this combination regimen.

Trial Registration: ClinicalTrials.gov Identifier: NCT05254847; registered prospectively on February 15, 2022.

Keywords: biliary tract cancer, adjuvant therapy, chemotherapy, immunotherapy, targeted therapy, prognosis

Background

Biliary tract cancer (BTC) is a rare but highly aggressive and fatal group of tumors, comprising three major subtypes: intrahepatic cholangiocarcinoma (ICC), extrahepatic cholangiocarcinoma [(ECC), including perihilar and distal cholangiocarcinoma], and gallbladder carcinoma (GBC).[1] These malignancies are typically diagnosed at an advanced stage, resulting in a poor prognosis with a 5-year survival rate of less than 5%.[2] Surgical resection offers the potential for curative treatment in early-stage BTC; however, fewer than 35% of cases are amenable to resection, and postoperative recurrence rates range from approximately 50% to 70%.[3] Consequently, there remains a critical unmet clinical need for effective strategies aimed at preventing recurrence and improving long-term survival outcomes.

Per-protocol analysis (PPS) of the BILCAP trial demonstrated that adjuvant capecitabine improved overall survival (OS) for resected BTC; however, this was not confirmed in the intention-to-treat (ITT) analysis.[4] Despite this, adjuvant capecitabine is now considered a standard of care, on the basis of the large effect size.[5] The phase III JCOG1202 ASCOT trial concluded that S-1 could be considered the standard of care for Asian patients.[6] However, most BTCs exhibit low sensitivity to chemotherapeutic agents or rapidly develop resistance.[7]

With advancements in immuno-oncology treatments, the combination of anti-angiogenesis and immunotherapy for this disease has garnered considerable interest.[8, 9] Mechanistically, programmed death-1 (PD-1) inhibitors and tyrosine kinase inhibitors (TKIs) enhance antitumor immune responses by promoting vascular normalization and improving the tumor microenvironment. Clinical evidence has demonstrated the feasibility and efficacy of postoperative adjuvant PD-1 inhibitors combined with targeted therapies, such as tislelizumab plus lenvatinib, in hepatocellular carcinoma.[10]

Tislelizumab, a novel humanized IgG4 monoclonal anti-PD-1 antibody, has been clinically demonstrated as a promising therapeutic option for various solid tumors.[11, 12] Although some clinicians have utilized tislelizumab monotherapy in the adjuvant setting for BTC, the extent of its clinical benefit remains uncertain. Lenvatinib, a multi-target tyrosine kinase inhibitor (TKI), has been extensively employed in the treatment of multiple malignancies due to its ability to inhibit diverse signaling pathways.[13] Evidence from studies indicates that combining lenvatinib with PD-1 inhibitors can synergistically enhance antitumor efficacy beyond what is achieved by either agent alone.[14] Several investigations have also provided data supporting the safety and clinical activity of lenvatinib combined with PD-1 inhibitors in patients with advanced BTC.[8, 9, 15] However, evidence regarding this adjuvant treatment strategy remains limited. A recently initiated randomized phase II trial (NCT03609489) is evaluating the efficacy and safety of adjuvant apatinib plus capecitabine compared to capecitabine monotherapy; currently, no published findings are available.

Building upon existing evidence, we conducted the first prospective study to evaluate the adjuvant use of a PD-1 inhibitor plus TKI and chemotherapy in patients with BTC following curative resection. This phase II study was designed to assess the efficacy and safety of a triple-agent regimen comprising tislelizumab, lenvatinib, and capecitabine. Herein, we present the clinical outcomes of this study. Additionally, exploratory analyses were performed to characterize the genomic landscape and identify potential biomarkers predictive of survival outcomes.

Methods

Study design

This prospective, single-center, single-arm, open-label phase II study was designed to evaluate the efficacy and safety of postoperative adjuvant therapy with capecitabine, lenvatinib, and tislelizumab in patients with BTC. The trial was prospectively registered on ClinicalTrials.gov (Identifier: NCT05254847) and conducted in strict accordance with the principles outlined in the Declaration of Helsinki and Good Clinical Practice guidelines. Ethical approval was obtained from the Ethics Committee of Fudan University Shanghai Cancer Center (Approval Number: 2109243-21). Written informed consent was obtained from all participants prior to enrollment.

Participants

Patients aged 18 years or older with histologically confirmed BTC, including ICC, ECC, or muscle-invasive GBC, were eligible. All patients were required to have undergone radical resection (R0), such as liver and/or pancreatic resection, and have an Eastern Cooperative Oncology Group (ECOG) performance status (PS) of 0-1. Other inclusion criteria included adequate organ function, a life expectancy of more than six months, and the ability to tolerate the planned treatment regimen.

Exclusion criteria included pancreatic cancer or mucosal GBC; prior chemoradiotherapy; incomplete postoperative recovery or unresolved biliary obstruction; confirmed distant metastases identified by imaging assessment; or concurrent malignancies. Patients presenting with abdominal fistulae, gastrointestinal perforation, or abdominal abscess within four weeks before study entry were also excluded. Additional exclusion criteria included active participation in other clinical trials or completion of a prior clinical trial within one month; patients who were expected or planned to receive any other systemic antitumor therapy during the study period; abnormal coagulation function; congenital or acquired immunodeficiency disorders (e.g., HIV infection or active hepatitis); and active autoimmune diseases or a documented history thereof.

Procedures

Patients received adjuvant therapy with tislelizumab, lenvatinib, and capecitabine within 4 to 16 weeks post-surgery. Tislelizumab was administered intravenously at a fixed dose of 200 mg on the first day of each 21-day cycle for a total of four cycles. Lenvatinib was administered orally at a daily dose of 8 mg for six months. Capecitabine was given orally at a dosage of 1250 mg/m² twice daily on days 1 to 14 within each 21-day cycle, for up to eight cycles. The treatment continued until either disease recurrence/metastasis was confirmed or intolerable toxicity occurred, whichever came first.

During the study period, dose reductions for tislelizumab were not permitted; permanent discontinuation with symptomatic management was mandated in cases of grade ≥ 3 adverse events (AEs). For lenvatinib, dose reduction, interruption, or discontinuation were implemented based on AE severity as defined by the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE), version 5.0. Adjustments to capecitabine dosing strictly adhered to established clinical practice guidelines.

Imaging assessments were performed by the investigator utilizing computed tomography (CT) or magnetic resonance imaging (MRI) in accordance with Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 at baseline and subsequently every nine weeks during therapy. Postoperative follow-up CT scans were scheduled at two-month intervals during the first year, every three months during the second year, and every six months thereafter. Safety monitoring was conducted throughout the treatment period and for 30 days post-treatment, encompassing evaluations of laboratory parameters including hematology, biochemistry, urinalysis, and coagulation profiles. Additionally, assessments included vital signs, physical examinations, ECOG PS, and 12-lead electrocardiograms. AEs were graded according to the NCI-CTCAE version 5.0.

Endpoints

The primary endpoint was the 1-year disease-free survival (DFS) rate. Secondary endpoints included DFS, 2-year DFS rate, OS, 1-year and 3-year OS rates, and safety. DFS was defined as the interval from the date of surgery to the first occurrence of recurrence, metastasis, or death from any cause, whichever occurred first. OS was defined as the time from diagnosis of BTC to death from any cause. The 1-year and 3-year OS rates represented the proportion of patients who survived within one or three years following a BTC diagnosis. Tumor recurrence was identified by radiological evidence indicating reappearance of tumors at any site post-surgery.

Exploratory analysis

A post-hoc exploratory analysis was conducted to investigate the correlation between genomic characteristics and clinical outcomes. Tumor tissues were collected from patients who had evaluable specimens and provided informed consent during surgical procedures for next-generation sequencing (NGS) analysis. Tissue samples with an estimated tumor purity below 10%, as determined by histopathological assessment, were considered insufficient for sequencing. The standard DNA input requirement was set at 250 ng, with a minimum threshold of 50 ng if DNA quality was suboptimal. No matched germline DNA analysis was performed for this exploratory analysis. Detailed methodologies for NGS testing have been described previously.[16]

Statistical analysis

The sample size was calculated based on the primary endpoint, which was the 1-year DFS rate. Referring to a previous trial[4] and our hypothesis that adjuvant combination therapy could improve the 1-year DFS rate in patients with BTC from 70% to 85%, a total of 67 patients was deemed sufficient to achieve a statistical power of 80% at a one-sided alpha level of 5%, assuming a recruitment period of 12 months and a follow-up duration of 36 months. Considering an anticipated dropout rate of approximately 10%, the final sample size was determined to be set at 75 patients.

Efficacy and safety were assessed in the as-treated population, defined as patients who received at least one dose of the investigational drug. Patient characteristics and AEs were summarized descriptively. DFS and OS were estimated using the Kaplan-Meier method with two-sided 95% confidence intervals (CIs). For DFS and OS analyses, data for patients who had not experienced relapse or death by the final follow-up were censored. Univariate and multivariate Cox proportional hazards regression models were employed to identify independent prognostic factors associated with DFS and OS. An exploratory analysis of DFS stratified by specific genetic mutations was conducted, with comparisons performed using a log-rank test to evaluate the prognostic significance of these mutations. All statistical tests were two-sided, with a significance level set at $P < 0.05$. Statistical analyses were conducted using SPSS software (version 26.0; SPSS Inc., IL, USA).

Results

Patient characteristics

Between February 24, 2022, and December 31, 2022, 91 patients were screened for eligibility and 65 patients were enrolled (**Figure 1; Additional file 1: Table S1**). Patient accrual was terminated early due to slow enrollment during the COVID-19 pandemic. Among 65 enrolled patients, 15 patients withdrew prior to initiating treatment due to loss to follow-up (n = 8) or personal decision (n = 7), and 50 patients received at least one dose of protocol therapy, including for efficacy and safety analyses (**Figure 1**). As of the data cutoff date (April 30, 2025), 38 patients had completed their assigned treatment regimen with no patient actively undergoing treatment. Twelve patients discontinued trial therapy mainly due to tumor recurrence (n = 8) and AEs (n = 2, one case each of hand-foot syndrome and ileus).

Patient baseline characteristics are summarized in **Table 1**. The median age of the patients was 58 years (range: 31-74), with 6 (12%) aged over 70 years and 29 (58%) being male. All patients had an ECOG PS of 0. As the study was conducted in a hepatic surgery department, patients with ICC accounted for 78% (39/50) of the cohort. No patients had received neoadjuvant therapy prior to enrollment. R0 resection was achieved in all cases. Most patients presented with pathological T stage 1 or 2 tumors [41/50 (82%)] and pathological TNM stage I or II disease [30/50 (60%)]. Additionally, well-differentiated or moderately differentiated tumors were identified in 20 patients (40%).

Efficacy

As of the data cutoff, the median follow-up duration was 29.3 months (95% CI: 27.2-30.4). Among the 50 patients with evaluable efficacy, 36 (72%) experienced disease recurrence. The median DFS was 12.7 months (95% CI: 6.4-19.0; **Figure 2a**), with a one-year DFS rate of 55.5% (95% CI: 51.4%-57.6%) and a two-year DFS rate of 30.8% (95% CI: 28.6%-32.9%). Data for OS were not mature at the time of data cutoff, with deaths reported in 20 patients (40%), primarily attributed to tumor recurrence and metastasis in 17 cases (85%) and hepatic failure in three cases (15%). The median OS had not been reached, while the one-year and three-year OS rates were recorded at 93.8% (95% CI: 91.7%-95.9%) and 54.4% (95% CI: 50.0%–58.4%), respectively (**Figure 2b**).

Given that the majority of enrolled patients were diagnosed with ICC (n=39 [78%]), patient characteristics (**Additional file 1: Table S1**) and efficacy were summarized for this subgroup. Among them, the median DFS was 14.6 months (95% CI: 6.5–22.6), with 1-year and 2-year DFS rates of 55.6% (95% CI: 53.0%–58.3%) and 31.8% (95% CI: 28.9%–34.4%), respectively. A total of 17 deaths occurred due to tumor recurrence or metastasis (n = 15 [88%]) and hepatic failure (n = 2 [12%]). The estimated OS rates were 94.7% (95% CI: 92.0%–97.4%) at one year and 50.1% (95% CI: 46.0%–54.7%) at three years.

The factors influencing DFS using Cox regression analyses in all patients are presented in **Table 2**. Univariate analysis identified pathological T stage and TNM stage as independent risk factors for DFS. Multivariate analysis further revealed that higher T stage (HR = 0.74; 95% CI: 0.29-0.99; P = 0.048) and advanced TNM stage (HR = 0.46; 95% CI: 0.21-0.97; P = 0.046) were significantly associated with poorer DFS outcomes (**Table 2; Figure 2c–2d**). Cox regression analyses identified no significant prognostic factors for OS; these analyses were exploratory since OS data remain immature (**Additional file 1: Table S2**).

Genomic landscape and biomarker analyses

In the post-hoc exploratory analysis, NGS was performed on 11 patients with evaluable specimens and informed consent; the genomic landscape is presented in **Additional file 1: Fig. S1a.** The mutation burden ranged from 3 to 17 mutations per tumor (median: 17; **Additional file 1: Fig. S1b**). The ten most frequently mutated genes in this cohort were *RAD50* (73%), *PRRT2* (64%), *MSH6* (55%), *BLM* (45%), *CEP290* (36%), *KRAS* (27%), *TP53* (27%), *BAP1* (18%), *HMCN1* (18%), and *FSIP2* (18%). Among the commonly mutated genes, *BAP1* and *FSIP2* mutations were exclusively observed in patients with disease recurrence (**Additional file 1: Fig. S1a**). We conducted exploratory analyses of associations between genomic alterations and clinical outcomes; these results are presented **Additional file 1: Fig. S2–S3**, but should be considered preliminary due to a small cohort without prespecified power. No commonly mutated genes were identified as being significantly associated with disease recurrence (**Additional file 1: Fig. S2**). Regarding DFS, patients harboring *FSIP2* mutations demonstrated shorter DFS compared to those without (P < 0.001); whereas no associations were observed for other commonly mutated genes (**Additional file 1: Fig. S3**).

Safety and tolerability

Safety data for the 50 as-treated patients are summarized in **Table 3**. As of the data cutoff, 43 out of 50 patients (86%) experienced at least one treatment-related adverse event (TRAE) as assessed by investigators. The most common TRAEs across all grades were hand-foot syndrome (28%), thrombocytopenia (24%), and elevated blood bilirubin levels (18%). The majority of TRAEs (66%) were classified as grade 1 or grade 2. Grade 3 TRAEs were reported in 10 patients (20%), including hand-foot syndrome (6%), elevated blood bilirubin levels (6%), rash (4%), lung infection (4%), and proteinuria (2%). During the study, dose reductions of lenvatinib due to TRAEs were required for nine patients (18%). Furthermore, two patients (4%) discontinued treatment due to hand-foot syndrome and ileus, with one case reported for each condition. No grade 4 TRAEs, serious adverse events, or treatment-related deaths occurred during the study period.

Discussion

To the best of our knowledge, this study is the first prospective analysis to evaluate the efficacy of a PD-1 inhibitor in combination with TKIs and chemotherapy as postoperative adjuvant therapy for BTC patients. Despite not meeting the primary endpoint, with 1-year DFS and OS rates of 55.5% and 93.8%, modest efficacy signals were observed with our triple regimen, particularly among patients with ICC (1-year DFS, 55.6%; 1-year OS, 94.7%), who derive limited benefit from adjuvant chemotherapy or radiotherapy.[17, 18] Given its manageable safety profile, further evaluation in larger, subtype-specific studies is warranted to better define its clinical utility.

Given the high recurrence rate and poor prognosis of BTC after surgical resection, adjuvant treatment has emerged as a reasonable therapeutic strategy. Regrettably, current adjuvant chemotherapy regimens, such as capecitabine, S-1, capecitabine plus gemcitabine, or gemcitabine plus platinum, have reported relapse rates ranging from 44% to 74%.[4, 6, 19, 20] Recent evidence indicates that combining chemotherapy with PD-1/PD-L1 inhibitors, with or without targeted agents, improves efficacy and maintains an acceptable safety profile in advanced BTC,[9, 15, 21, 22] suggesting that such combination approaches may also hold promise in the adjuvant setting.

Our study reports a relapse rate of 72%. The median DFS of 12.7 months observed in our study appears less favorable when compared to other studies, such as the BILCAP and ASCOT trials.[4, 6, 19, 20]

However, it is important to emphasize that the patient populations in the BILCAP and ASCOT studies differed significantly from ours in terms of disease distribution. Specifically, only 19% and 14% of their patients were diagnosed with ICC, whereas ICC accounted for 78% of cases in our cohort. Subgroup analyses from the BILCAP study suggest that patients with ECC may derive greater benefit from adjuvant chemotherapy.[23] On the contrary, ICC demonstrates a relatively limited response to standard chemotherapy following surgery due to high drug resistance and low treatment sensitivity.[17, 18] Additionally, differences in tumor staging and differentiation between these studies further contextualize our findings. In the BILCAP trial, 87% of patients treated with capecitabine had tumors classified as TNM stage I–II; similarly, this proportion was reported as 75% among patients treated with S-1 in the ASCOT trial. By contrast, only 60% of our study population had TNM stage I-II tumors. Moreover, while well- or moderately differentiated tumors comprised approximately 64% and 82% of cases in the BILCAP and ASCOT trials respectively; this proportion was markedly lower at just 40% within our cohort (see **Supplemental Table S3**).[4, 6] Both TNM staging and tumor differentiation are widely recognized as critical prognostic factors,[17, 24] with TNM staging also demonstrating prognostic significance in our subgroup analysis. These distinctions likely contribute to explaining why DFS outcomes observed in our study were inferior compared to those reported by prior investigations.

Although the OS data from our study are still immature and should be interpreted with caution, we observed estimated 1-year and 3-year OS rates of 93.8% and 54.4% at a median follow-up of 29.3 months, which were numerically similar to those reported for established chemotherapy regimens in the BILCAP (ITT, 87.4% and 52.9%; PPS, 90.5% and 56.2%)[23] and ASCOT (ITT, 95.9% and 76.2%)[6] trials. Of particular note, both the BILCAP (HR, 1.12; 95% CI: 0.79–1.60) and ASCOT (HR, 0.61; 95% CI: 0.30–1.23) trials showed no significant OS benefit from capecitabine or S-1 among the sub-population with ICC. [6, 23] Consistent with these findings, retrospective analysis of a large ICC cohort (n = 1159) also failed to demonstrate a clear OS advantage with adjuvant chemotherapy.[25] Collectively, these results underscore the ongoing uncertainty regarding the efficacy of adjuvant chemotherapy specifically for ICC, despite their broader clinical endorsement in BTC. Given that 78% of patients in our study cohort were diagnosed with ICC, the observed OS outcomes may offer preliminary evidence supporting further investigation of combinatorial adjuvant regimens in BTC—particularly for ICC, where current treatment options remain suboptimal. Overall, given the inherent limitations of cross-trial comparisons, along with variations in study design, treatment regimens, and patient characteristics, definitive conclusions cannot yet be drawn and should be validated in larger studies with extended follow-up.

In the exploratory analysis, the genomic landscape observed in our study largely overlapped with previously reported mutational profiles, including frequent alterations in *KRAS* and *TP53*.[23, 26, 27] Among these, *FSIP2* mutation appeared to be associated with shorter DFS, consistent with prior observations in other malignancies, such as renal cell carcinoma and gastric cancer.[27, 28] These findings may suggest a potential biological relevance of *FSIP2* in BTC. However, given the very limited sample size and post-hoc nature of this analysis, these findings should be interpreted with caution and regarded as hypothesis-generating. Further validation in larger, independent cohorts is warranted.

The combination of tislelizumab, lenvatinib, and capecitabine demonstrated a manageable safety profile, with no new safety signals identified in patients who had undergone curative resection for BTC.

Among the 50 enrolled patients, 76% (38/50) successfully completed the full course of scheduled treatment, while only 4% (2/50) discontinued therapy due to TRAEs. The overall safety profile was consistent with that observed for each agent administered as monotherapy, indicating that the triplet regimen did not lead to unexpected toxicities or heightened safety concerns.[4, 29-31] Although most patients experienced at least one TRAE, the majority of AEs were grade 1 or 2 in severity and were effectively managed through supportive care or dose modifications. Treatment discontinuation due to TRAEs was rare (4%), further underscoring the tolerability of this regimen. While direct comparisons with historical chemotherapy regimens are limited by differences in study design and patient populations, the favorable toxicity profile observed across multiple tumor types suggests potential clinical applicability.[32, 33] Nonetheless, the incidence of AEs reported in this study may be underestimated due to factors such as the relatively small sample size, younger median age, and favorable baseline performance status of the cohort. Therefore, prospective randomized trials with larger sample sizes are necessary to validate both the safety and therapeutic efficacy of this combination in an adjuvant setting.

Several limitations in our study warrant consideration. Firstly, the primary limitation was the early termination of our study due to slow enrollment during the COVID-19 pandemic. Secondly, this exploratory phase 2 study employed an open-label, single-arm, non-randomized design, which inherently lacks a direct comparison with standard-of-care treatments or other established regimens. Thirdly, the trial was conducted in a single center with most patients with ICC, which may reflect institutional referral patterns rather than the true distribution of BTC subtypes, potentially restricting the generalizability to broader populations. Additionally, due to the relatively short follow-up period at the time of analysis, survival data remain immature and insufficient for assessing long-term benefits. Extended follow-up is required to obtain more robust survival outcomes. Finally, the lack of pre-specified biomarker analyses, particularly PD-L1 expression, limited available samples, and the absence of matched germline DNA analysis also restricted the comprehensive and accurate assessment of predictive biomarkers.

Conclusions

In summary, this study suggested the feasibility and acceptable tolerability of adjuvant tislelizumab, lenvatinib, and capecitabine in patients with BTC, particularly in those with ICC, although the predefined efficacy threshold was not met. Furthermore, the exploratory analysis identified the *FSIP2* mutation as a potential prognostic biomarker, underscoring the importance of comprehensive genomic profiling. Future multicenter, large-scale studies in selected populations incorporating integrated molecular analyses are warranted to validate and extend these results.

List of abbreviations

BTC, biliary tract cancer;

ICC, intrahepatic cholangiocarcinoma;

ECC, extrahepatic cholangiocarcinoma;

PPS, per-protocol analysis;

OS, overall survival;

ITT, intention-to-treat;

GBC, gallbladder carcinoma;

PD-1, programmed death-1;

TKI, tyrosine kinase inhibitors;
ECOG PS, Eastern Cooperative Oncology Group performance status;
AEs, adverse events;
NCI-CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events;
CT, computed tomography;
MRI, magnetic resonance imaging;
DFS, disease-free survival;
NGS, next-generation sequencing;
CIs, confidence intervals;
TRAE: treatment-related adverse event.

Declarations

Ethics approval and consent to participate

This study was approved by the Ethics Committee of Fudan University Shanghai Cancer Center (approval number: 2109243-21). All procedures conducted in this study adhered to the ethical standards of the institutional and/or national research committee, Good Clinical Practice guidelines, as well as the Declaration of Helsinki and its subsequent amendments or equivalent ethical standards. Written informed consent was obtained from all patients.

Consent for publication

All patients provided informed consent for the publication of any associated data.

Data Availability

The data supporting the finding, trial protocol and statistical analysis plan of this study can be obtained from the corresponding author upon reasonable request.

Competing interests

The authors have no conflicts of interest to declare.

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Authors' contributions

YF: Data curation, formal analysis, investigation, methodology, software, visualization, writing-original draft; NZ: Data curation, formal analysis, investigation, methodology; YMZ: Resources; QP: Resources; ARM: Resources; WPZ: Resources; TZ: Conceptualization, funding acquisition, methodology, project administration, resources, supervision, validation, writing-review and editing; LW: Conceptualization, funding acquisition, methodology, project administration, resources supervision validation writing-review and editing. All authors read and approved the final manuscript.

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References

1. Vogel A, Bridgewater J, Edeline J, Kelley RK, Klumpen HJ, Malka D, et al. Biliary tract cancer: ESMO Clinical Practice Guideline for diagnosis, treatment and follow-up. Ann Oncol. 2023;34:127-40.
2. Marin JJG, Prete MG, Lamarca A, Tavolari S, Landa-Magdalena A, Brandi G, et al. Current and novel therapeutic opportunities for systemic therapy in biliary cancer. Br J Cancer. 2020;123:1047-59.
3. Lamarca A, Edeline J, McNamara MG, Hubner RA, Nagino M, Bridgewater J, et al. Current standards and future perspectives in adjuvant treatment for biliary tract cancers. Cancer Treat Rev. 2020;84:101936.
4. Primrose JN, Fox RP, Palmer DH, Malik HZ, Prasad R, Mirza D, et al. Capecitabine compared with observation in resected biliary tract cancer (BILCAP): a randomised, controlled, multicentre, phase 3 study. Lancet Oncol. 2019;20:663-73.
5. Shroff RT, Kennedy EB, Bachini M, Bekaii-Saab T, Crane C, Edeline J, et al. Adjuvant Therapy for Resected Biliary Tract Cancer: ASCO Clinical Practice Guideline. J Clin Oncol. 2019;37:1015-27.
6. Nakachi K, Ikeda M, Konishi M, Nomura S, Katayama H, Kataoka T, et al. Adjuvant S-1 compared with observation in resected biliary tract cancer (JCOG1202, ASCOT): a multicentre, open-label, randomised, controlled, phase 3 trial. Lancet. 2023;401:195-203.
7. Sohal DP, Shrotriya S, Abazeed M, Cruise M, Khorana A. Molecular characteristics of biliary tract cancer. Crit Rev Oncol Hematol. 2016;107:111-8.
8. Xie L, Huang J, Wang L, Ren W, Tian H, Hu A, et al. Lenvatinib Combined With a PD-1 Inhibitor as Effective Therapy for Advanced Intrahepatic Cholangiocarcinoma. Front Pharmacol. 2022;13:894407.
9. Zhu C, Xue J, Wang Y, Wang S, Zhang N, Wang Y, et al. Efficacy and safety of lenvatinib combined with PD-1/PD-L1 inhibitors plus Gemox chemotherapy in advanced biliary tract cancer. Front Immunol. 2023;14:1109292.
10. Yang J, Jiang S, Chen Y, Zhang J, Deng Y. Adjuvant ICIs Plus Targeted Therapies Reduce HCC Recurrence after Hepatectomy in Patients with High Risk of Recurrence. Curr Oncol. 2023;30:1708-19.
11. Shen L, Guo J, Zhang Q, Pan H, Yuan Y, Bai Y, et al. Tislelizumab in Chinese patients with advanced solid tumors: an open-label, non-comparative, phase 1/2 study. J Immunother Cancer. 2020;8.
12. Xu J, Bai Y, Xu N, Li E, Wang B, Wang J, et al. Tislelizumab Plus Chemotherapy as First-line Treatment for Advanced Esophageal Squamous Cell Carcinoma and Gastric/Gastroesophageal Junction Adenocarcinoma. Clin Cancer Res. 2020;26:4542-50.
13. Hao Z, Wang P. Lenvatinib in Management of Solid Tumors. Oncologist. 2020;25:e302-e10.
14. Kimura T, Kato Y, Ozawa Y, Kodama K, Ito J, Ichikawa K, et al. Immunomodulatory activity of lenvatinib contributes to antitumor activity in the Hepa1-6 hepatocellular carcinoma model. Cancer Sci. 2018;109:3993-4002.
15. Shi GM, Huang XY, Wu D, Sun HC, Liang F, Ji Y, et al. Toripalimab combined with lenvatinib and GEMOX is a promising regimen as first-line treatment for advanced intrahepatic

cholangiocarcinoma: a single-center, single-arm, phase 2 study. Signal Transduct Target Ther. 2023;8:106.

16. Wang L, Zhu H, Zhao Y, Pan Q, Mao A, Zhu W, et al. Comprehensive molecular profiling of intrahepatic cholangiocarcinoma in the Chinese population and therapeutic experience. J Transl Med. 2020;18:273.
17. Wilbur HC, Soares HP, Azad NS. Neoadjuvant and adjuvant therapy for biliary tract cancer: Advances and limitations. Hepatology. 2025;82:1287–302.
18. Kefas J, Bridgewater J, Vogel A, Stein A, Primrose J. Adjuvant therapy of biliary tract cancers. Ther Adv Med Oncol. 2023;15:17588359231163785.
19. Ben-Josef E, Guthrie KA, El-Khoueiry AB, Corless CL, Zalupski MM, Lowy AM, et al. SWOG S0809: A Phase II Intergroup Trial of Adjuvant Capecitabine and Gemcitabine Followed by Radiotherapy and Concurrent Capecitabine in Extrahepatic Cholangiocarcinoma and Gallbladder Carcinoma. J Clin Oncol. 2015;33:2617-22.
20. Edeline J, Benabdelghani M, Bertaut A, Watelet J, Hammel P, Joly JP, et al. Gemcitabine and Oxaliplatin Chemotherapy or Surveillance in Resected Biliary Tract Cancer (PRODIGE 12-ACCORD 18-UNICANCER GI): A Randomized Phase III Study. J Clin Oncol. 2019;37:658-67.
21. Oh DY, He AR, Bouattour M, Okusaka T, Qin S, Chen LT, et al. Durvalumab or placebo plus gemcitabine and cisplatin in participants with advanced biliary tract cancer (TOPAZ-1): updated overall survival from a randomised phase 3 study. Lancet Gastroenterol Hepatol. 2024;9:694-704.
22. Kelley RK, Ueno M, Yoo C, Finn RS, Furuse J, Ren Z, et al. Pembrolizumab in combination with gemcitabine and cisplatin compared with gemcitabine and cisplatin alone for patients with advanced biliary tract cancer (KEYNOTE-966): a randomised, double-blind, placebo-controlled, phase 3 trial. Lancet. 2023;401:1853-65.
23. Bridgewater J, Fletcher P, Palmer DH, Malik HZ, Prasad R, Mirza D, et al. Long-Term Outcomes and Exploratory Analyses of the Randomized Phase III BILCAP Study. J Clin Oncol. 2022;40:2048-57.
24. Peng Y, Liang A, Chen Z, Yang B, Yu W, Deng J, et al. The efficacy of adjuvant chemotherapy for curative resected biliary tract cancers: a systematic review and network meta-analysis of randomized clinical trials. Int J Surg. 2025;111:2182-94.
25. Sharib J, Rhodin KE, Liu A, McIntyre S, Bartholomew A, Masoud S, et al. Adjuvant Cytotoxic Chemotherapy may not be Associated with a Survival Advantage for Resected Intrahepatic Cholangiocarcinoma. Ann Surg Oncol. 2025;32:2456-66.
26. Lin J, Cao Y, Yang X, Li G, Shi Y, Wang D, et al. Mutational spectrum and precision oncology for biliary tract carcinoma. Theranostics. 2021;11:4585-98.
27. Guo L, Zhou F, Liu H, Kou X, Zhang H, Chen X, et al. Genomic mutation characteristics and prognosis of biliary tract cancer. Am J Transl Res. 2022;14:4990-5002.
28. Zhang Y, Zhu X, Qiao X, Sun L, Tian Y, Yang Y, et al. FSIP2 can serve as a predictive biomarker for Clear Cell Renal Cell Carcinoma prognosis. Int J Med Sci. 2020;17:2819-25.
29. Ueno M, Ikeda M, Sasaki T, Nagashima F, Mizuno N, Shimizu S, et al. Phase 2 study of lenvatinib monotherapy as second-line treatment in unresectable biliary tract cancer: primary analysis results. BMC Cancer. 2020;20:1105.

30. Zhou C, Huang D, Fan Y, Yu X, Liu Y, Shu Y, et al. Tislelizumab Versus Docetaxel in Patients With Previously Treated Advanced NSCLC (RATIONALE-303): A Phase 3, Open-Label, Randomized Controlled Trial. J Thorac Oncol. 2023;18:93-105.
31. Desai J, Deva S, Lee JS, Lin CC, Yen CJ, Chao Y, et al. Phase IA/IB study of single-agent tislelizumab, an investigational anti-PD-1 antibody, in solid tumors. J Immunother Cancer. 2020;8: e000453.
32. Xu L, Chen J, Liu C, Song X, Zhang Y, Zhao H, et al. Efficacy and safety of tislelizumab plus lenvatinib as first-line treatment in patients with unresectable hepatocellular carcinoma: a multicenter, single-arm, phase 2 trial. BMC medicine. 2024;22:172.
33. Tan K, He X, Yuan H, Zhao S, Gong C, Zhao Y, et al. Evaluating Tislelizumab, Lenvatinib, and FOLFOX4-HAIC as a Conversion Therapy for Unresectable Hepatocellular Carcinoma. ILIVER. 2023;2:163-9.

Figures legends

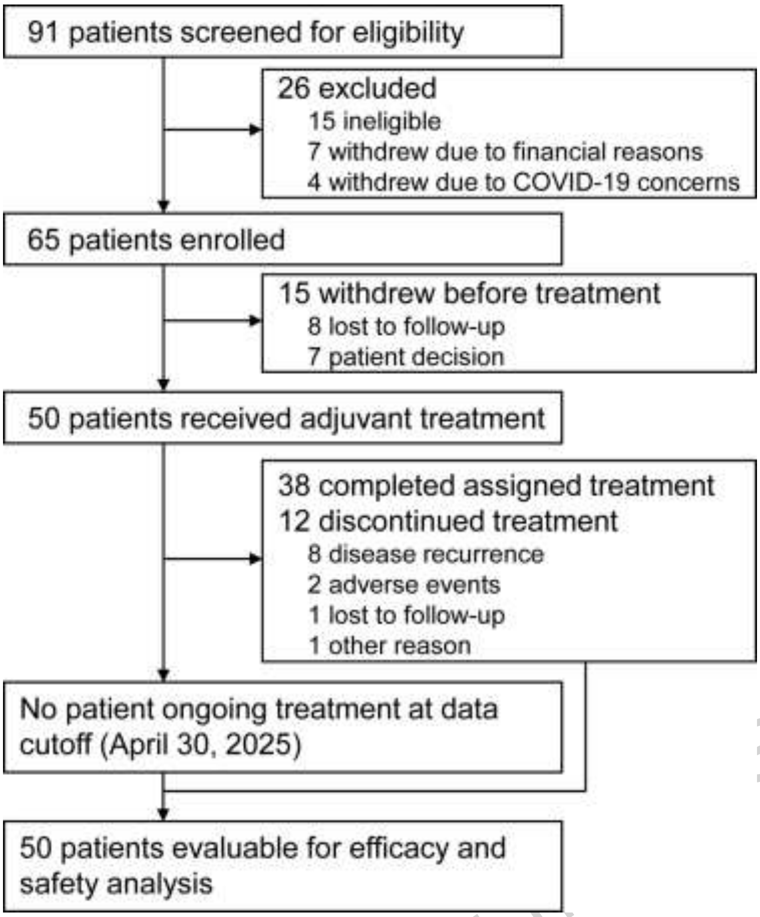

Figure 1. Trial profile.

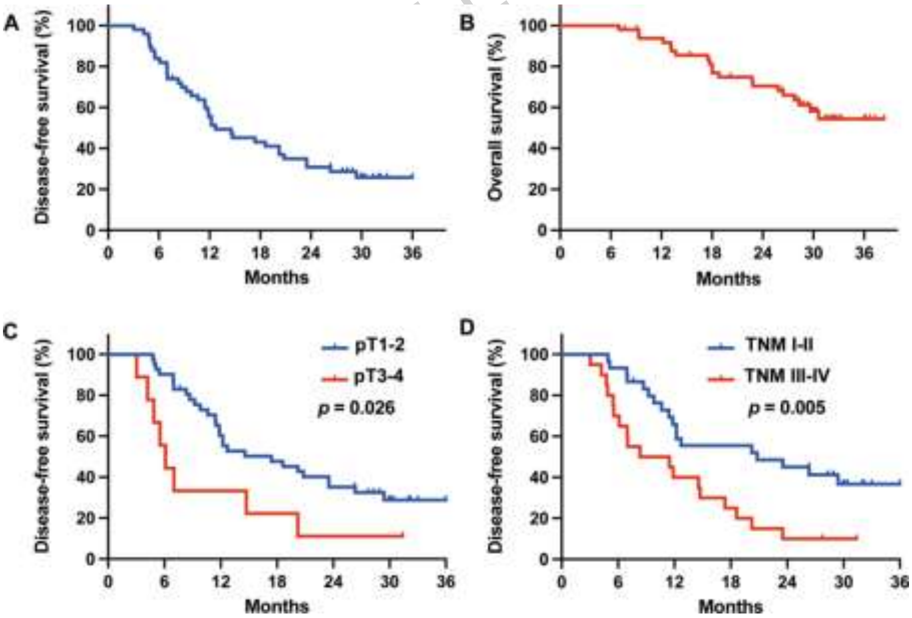

Figure 2. Kaplan-Meier curves of outcomes. (A) Disease-free survival. (B) Overall survival. (D) Disease-free survival stratified by tumor T stage. (D) Disease-free survival stratified by tumor TNM stage.

Appendix File 1: Tables S1-S3 and Figures S1-S3.

Appendix Table S1. Baseline characteristics of all enrolled patients and ICC patients

Appendix Table S2. Univariate and multivariate Cox regression analyses for overall survival in all treated patients

Appendix Table S3. Comparison of the general characteristics of patients in the current phase 3 clinical trials with positive results and this study.

Appendix Figure S1. Genomic landscape and mutation type overview.

Appendix Figure S2. Relapse rates stratified by mutation status of commonly mutated genes. Fisher's exact test was employed to assess statistical significance between the two subgroups.

Appendix Figure S3. Disease-free survival stratified by mutation status of commonly mutated genes. Statistical significance between the two subgroups was assessed using the log-rank test.